Discrete random variable probability distributions

1 State whether the following tables represent a discrete probability distribution. Explain your answer.

a

x	2	4	5	6	7
$P(X = x)$	0.1	0.25	0.3	0.15	0.2

b

x	-2	-1	4	0	3
$P(X = x)$	-0.4	0.05	0.15	0.2	0

2 State whether the following tables represent a discrete probability distribution:

a

x	2	4	6	8
$p(x)$	0.2	0.4	0.6	0.8

b

x	2	4	6	8	10
$p(x)$	0.2	0.2	0.2	0.2	0.2

c

x	1	2	3	4
$p(x)$	-0.4	-0.3	0.8	0.9

d

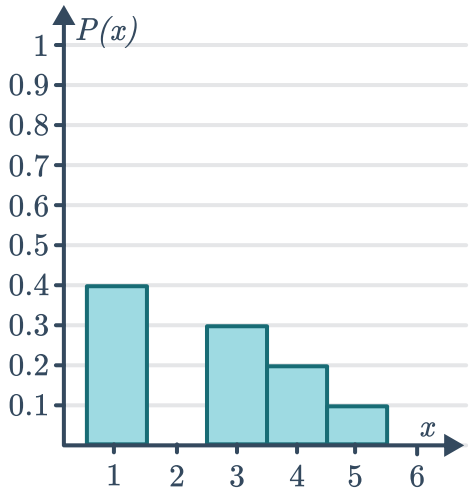
x	10	20	30	40
$p(x)$	10%	20%	25%	45%

3 For each of the the following column graphs:

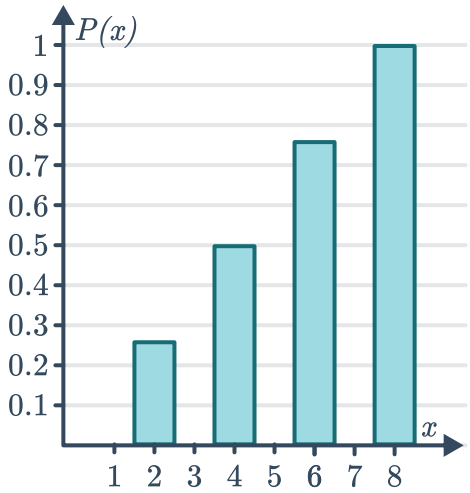
i Identify which conditions for a discrete probability distribution are evident in the graph.

ii Hence, state whether the graph represents a discrete probability distribution.

a

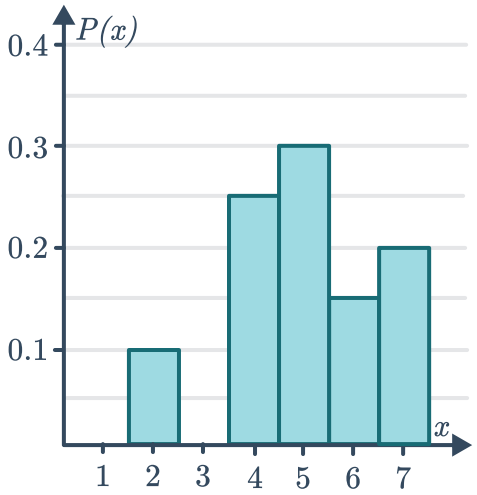


b

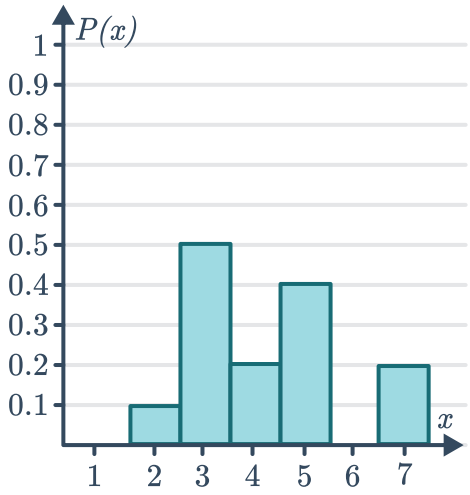


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c



d



4 Consider the function $P(X = x) = \frac{x}{6}$ for $x = 1, 2, 3$.

a Complete the given table:

b State whether the table represents a discrete probability distribution. Explain your answer.

x	1	2	3
$P(X = x)$			

5 Consider the function $P(X = x) = -\frac{x}{3}$ for $x = 1, 2, 3$.

a Complete the given table:

b State whether the table represents a discrete probability distribution. Explain your answer.

x	1	2	3
$P(X = x)$			

6 Consider the function $P(X = x) = \frac{1}{8}x^2$ for $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$.

a Complete the given table:

b State whether the table represents a discrete probability distribution. Explain your answer.

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
$P(X = x)$					

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7 A random variable X has the following probability distribution:

x	5	6	7	8	9	12
$P(X=x)$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{3}{16}$	$\frac{1}{4}$	$\frac{5}{16}$	$\frac{1}{8}$

Find the following probabilities:

a $P(X=8)$ **b** $P(X \text{ is even})$ **c** $P(X>8)$ **d** $P(X \leq 7)$

e $P(6<X<8)$ **f** $P(6 \leq X<12)$

8 The cumulative distribution for a discrete random variable is given in the below:

a Complete the given table.

b Find the following probabilities:

i $P(1<X \leq 3)$ ii $P(X \leq 3|X>1)$

x	0	1	2	3	4
$P(X \leq x)$	0.1	0.3	0.65	0.95	1
$P(X=x)$					

9 The cumulative distribution for a discrete random variable is given in the below:

a Complete the given table.

b Find the following probabilities correct to three decimal places.

i $P(X<4)$

ii $P(X \geq 8)$

iii $P(X>8 \cup X<3)$

iv $P(X \leq 7|X \geq 5)$

x	$P(X \leq x)$	$P(X=x)$
0	0.001	
1	0.011	
2	0.055	
3	0.172	
4	0.377	
5	0.623	
6	0.828	
7	0.945	
8	0.989	
9	0.999	
10	1	

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10 A random variable X has the following probability distribution:

x	1	2	3	4	5	6
$P(X=x)$	$\frac{5}{60}$	$\frac{7}{60}$	$\frac{9}{60}$	$\frac{11}{60}$	$\frac{13}{60}$	$\frac{15}{60}$

The probability density function associated with this distribution is given by:

$$P(X=x)=\frac{ax+b}{c}$$

State the value of:

a c **b** a **c** b

11 The following tables represent the probability distribution of a discrete random variable for all of the possible outcomes. For each table, determine the value of k:

a

x	-3	-1	0	1	4
$P(X=x)$	0.3	k	0.05	0.2	0.25

b

x	-2.8	-0.7	0.2	1	3.4
$P(X=x)$	0.3	0.1	0.05	k	0.25

12 Given the probability distribution table for X below, find each of the following:

x	0	1	2	3	4
$P(X=x)$	$2k$	k	$4k$	$3k$	k

a The value of k **b** $P(X>2)$

c $P(1\leq X<4)$ **d** $P(X>2|X>0)$

e $P(X\leq 3|X\geq 1)$

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13 For each of the probability functions for a discrete random variable, described below, determine the value of k:

a $$P(X=x)=\begin{cases}\frac{k}{x};\ x=2,3,4,5,6,10\\ 0,\ \text{otherwise}\end{cases}$$

b $$P(X=x)=\begin{cases}kx^2(5-x);\ x=1,2,3,4,5\\ 0,\ \text{otherwise}\end{cases}$$

c $$P(X=x)=\begin{cases}\frac{x^2}{k};\ x=-3,-2,-1,0,1,2\\ 0,\ \text{otherwise}\end{cases}$$

d $$P(X=x)=\begin{cases}\frac{kx}{24};\ x=\frac{1}{8},\frac{1}{6},2,4,6,8\\ 0,\ \text{otherwise}\end{cases}$$

14 A random variable X can take any of the values 0, 1, 2 or 3. Given the following known facts about the distribution, construct a probability distribution table for each case below:

a

- $P(X=0)=P(X=1)$
- $P(X=1)=P(X=2)$
- $P(X=2)=P(X=3)$

b

- $P(X=0)=\frac{1}{2}P(X=1)$
- $P(X=1)=4P(X=2)$
- $P(X=2)=\frac{1}{5}P(X=3)$

c

- $P(X=2)=2P(X=0)$
- $P(X=1)=P(X=2)$
- $P(X=3)=4P(X=1)$

d

- $P(X=3)=3P(X=0)$
- $P(X=1)=5P(X=0)$
- $P(X=1)+P(X=2)=\frac{5}{7}$

15 Consider the following probability distribution table for X:

x	0	1	2	3	4
$P(X=x)$	0.24	a	0.15	b	0.22

a Find the value of a given that $P(X<3)=0.62$.

b Hence, determine the value of b.

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16 Consider the following probability distribution table for X:

x	1	2	3	4	5
$P(X=x)$	0.16	a	b	c	0.17

a Find the value of c given that $P(X \geq 4) = 0.39$.

b Find the value of a given that $P(X \geq 2 | X < 3) = 0.6$.

c Hence, determine the value of b.

17 The probability function for a discrete random variable is given by:

$$P(X=x) = \begin{cases} k(9-x);\ x = 4,5,6,7,8 \\ 0,\ \text{otherwise} \end{cases}$$

a Determine the value of k.

b Construct a probability distribution table.

c Find $P(X < 7)$.

d Find $P(X \geq 6)$.

e Find $P(X < 6 \cup X > 7)$.

f Find $P(X \geq 5 | X \leq 7)$.

18 The probability function for a discrete random variable is given by:

$$P(X=x) = \begin{cases} {}^5C_x\,(0.6)^x\,(0.4)^{5-x};\ x = 0,1,2,3,4,5 \\ 0,\ \text{for all other values of } x \end{cases}$$

a Construct a probability distribution table.

b Describe the shape of the distribution.

c Find $P(X > 3)$.

d Find $P(X > 0)$.

e Find $P(X \leq 3 | X > 0)$.

19 The probability function for a discrete random variable is given by:

$$P(X=x) = \left(\frac{1}{2}\right)^x;\ x = 1,2,3,\ldots$$

a State whether the probabilities of this distribution form a geometric or arithmetic sequence.

b State the common difference or ratio of this sequence.

c State the first term of this sequence.

d Find $\sum_{x=0}^{\infty} P(X=x)$, to confirm $P(X=x)$ describes a probability function..

e Find $P(2 < X \leq 5)$.

f Find $P(X \leq 7 | X < 10)$.

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Uniform distribution

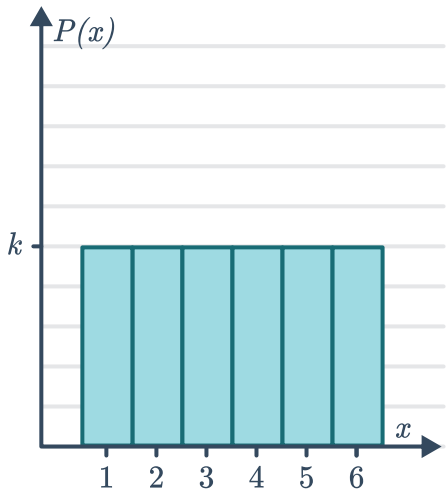
20 Consider the function $P(X = x) = \frac{1}{6}$ for $x = 0, 2, 4, 6, 8, 10$.

a Complete the following table:

b State whether the table represents a discrete probability distribution. Explain your answer.

x	0	2	4	6	8	10
$P(X = x)$						

21 The probability function for a uniform discrete random variable is shown in the following graph:



a Find the value of k.

b State the probability function that defines the uniform discrete random variable representated by the graph.

22 Given the probability distribution table for X below, find each of the following:

a The value of k.

b $P(X > 3)$

c $P(1 \leq X < 5)$

d $P(X > 2 | X > 1)$

e $P(X \leq 4 | X \geq 2)$

x	1	2	3	4	5
$P(X = x)$	k	k	k	k	k

23 The probability function for a uniform discrete random variable is given below:

$$P(X = x) = \begin{cases} k; x = 1, 2, 3, 4 \\ 0, \text{ otherwise} \end{cases}$$

a Find the value of k.

b Find $P(X < 3)$.

c Find $P(X \geq 2 | X < 4)$.

d Find m such that $P(X \geq m) = 0.75$.

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Applications

24 A coin is weighted such that the probability of a tail appearing uppermost is 70%. Let X represent the number of tails appearing uppermost in two tosses of the coin.

a Construct a tree diagram for this situation.

b Hence, complete the table:

x	0	1	2
$P(X=x)$			

c State the three conditions that must be true for this to be a discrete probability distribution.

d Hence, does this represent a discrete probability distribution?

25 A dog has three puppies. Let F represent the number of female puppies in this litter.

a Represent all possible outcomes of male and female puppies born using a tree diagram.

b Construct the probability distribution table for F.

c Is the discrete probability distribution uniform or non-uniform?

26 A normal six-sided die is rolled 6000 times.

a List the possible outcomes of each roll.

b What shape do you expect the distribution resulting from the experiment to be? Explain your answer.

c The results of the experiment are tabulated below:

x	1	2	3	4	5	6
Frequency	996	1005	994	1005	996	1004

Hence, complete the table below, leaving your answer as a fraction over 6000:

x	1	2	3	4	5	6
$p(x)$						

d Using the results, calculate the experimental probability $p(X<3)$.

e Find the percentage margin of error between the experimental and theoretical value of $P(X<3)$.

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27 Consider a normal and fair eight-sided die. We are interested in how long it takes for a 5 to appear uppermost on the die for the first time.

a State the probability of:

i A 5 appearing uppermost on the first roll of the die.

ii A 5 first appearing uppermost on the third roll of the die.

iii A 5 first appearing uppermost on the fourth roll of the die.

iv A 5 first appearing uppermost on the seventh roll of the die.

b Let X be the number of rolls of the die required to see a 5 for the first time. State the conditions that are required for a discrete probability distribution for X.

c Hence, does this represent a discrete probability distribution?

28 A fair six-sided die is rolled. Let X be the number it lands on.

a Construct a probability distribution table for X.

b Find $P(X \geq 5)$.

c Find $P(X \geq 5 | X \leq 5)$.

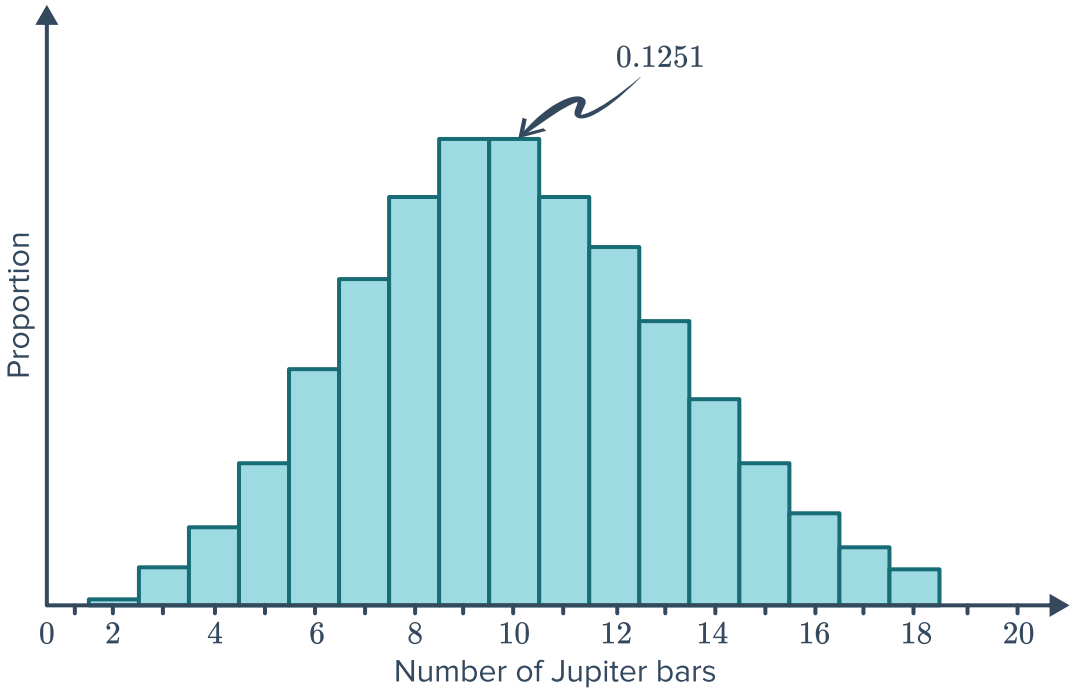
29 A six-sided die is weighted such that the probability of the die landing with any of the numbers on the die facing up is directly proportional to that number. For example, the probability of the die landing with 5 facing up, is $5k$, where k is a positive constant.

a Find the value of k.

b Let X be the number appearing facing upwards. Construct the probability distribution table for X.

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30 In a mixed Bradbury chocolate showbag, Bradbury claims that each bag contains on average 10 Jupiter bars. The showbag committee tested thousands of bags and the graph below shows the relative frequency of Jupiter bars in the Bradbury bags:



a State the probability that there are 10 Jupiter bars in a Bradbury showbag.

b Does it appear that Bradbury's claim about the number of Jupiter bars is correct? Explain your answer.

c The showbag committee observe that their graph does indeed represent a discrete random variable and can be modelled by:

$$P(X = x) = \frac{10^x e^{-10}}{x!}$$

Use the formula to find the following probabilities, rounding your answers to four decimal places:

i $P(X = 12)$ **ii** $P(X = 7)$

iii $P(X \leq 3)$ **iv** $P(4 \leq X \leq 6)$

d Bradbury also claim that over 50% of customers will receive between 8 and 12(inclusive) bars in a showbag. Is this claim correct? Give supporting evidence.

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31 Several boxes of fruit delivered to an office building were sampled. It was found that the number of fruit in each box wasn't always the same. The frequency of each observation is given in the table:

Number of fruit	Frequency
38	5
39	7
40	24
41	8
42	26

a Let X be the number of fruit in a box. Construct a probability distribution table for X using the observations.

b Use the experimental probabilities to estimate the following:

i The probability that the next box delivered contains at least 40 fruit.

ii The probability that the next box delivered contains less than 41 fruit.

32 NetStan, an on-demand internet streaming provider, surveyed their customers asking them which TV programme convinced them to subscribe to NetStan. The results are summarised in the table on the right:
Does this table represent a discrete probability distribution? Explain your answer.

Response	$P(X = x)$
The Old Cafe	0.31
Deadliest Animals Alive	0.12
Attack of the Killer Lawnchairs	0.27
The Big Rash	0.3

33 The probability of being placed in a queue when calling your electricity provider is given by:

$$p(x) = k \times (0.2)^x \; ; \; x = 0, 1, 2, 3, \ldots$$

Where x is how many customers are in the queue before you, and k is a positive constant.

a Given the probability of there being 2 people in the queue before you is 0.032, find the value of k.

b Find the probability that you are not placed in a queue at all.

c Find the probability of you being 6th in line to be served.

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